

SYMPLECTIC, POISSON, AND CONTACT KELLER COUNTEREXAMPLES FROM ALPÖGE’S MAP, WITH MACHINE-CHECKED CERTIFICATION

MARCUS TELLER

ABSTRACT. Alpöge’s counterexample to the Jacobian conjecture in dimension three (announced July 19, 2026, US time) has immediate consequences beyond the Jacobian problem. We work three of them out explicitly and certify every computational claim by machine-checked Lean 4 proofs. First, the cotangent lift of the map is an injective, non-surjective endomorphism of the third Weyl algebra A_3 , disproving the Dixmier conjecture DC_3 ; an independent note of Pickhardt and an AI research agent [12] carries out the same transfer, and our treatment adds the certification layer. Second, the classical shadow of that lift is a polynomial endomorphism of $\mathbb{C}^6$ preserving the canonical Poisson bracket, with Jacobian determinant identically 1 and not injective: an explicit counterexample to the Poisson conjecture PC_3 and, to our knowledge, the first symplectic Keller counterexample. Third, the lift preserves the Liouville form on the nose, so its contactization is a polynomial map of $\mathbb{C}^7$ preserving the standard contact form exactly, with Jacobian determinant 1 and not injective: the first non-invertible polynomial strict contactomorphism of affine space, answering (negatively) the natural contact analogue of the Jacobian conjecture, which we formulate. By stabilization all three phenomena persist in every higher dimension. The cases $n \in \{1, 2\}$ of the Dixmier and Poisson conjectures remain open; for $n = 1$ our construction sharpens the question of which plane Keller maps quantize.

1. INTRODUCTION

Let $A_n = \mathbb{C}\langle \hat{x}_1, \dots, \hat{x}_n, \partial_1, \dots, \partial_n \rangle$ denote the n -th Weyl algebra, with defining relations $[\partial_i, \hat{x}_j] = \delta_{ij}$, $[\hat{x}_i, \hat{x}_j] = [\partial_i, \partial_j] = 0$. The *Dixmier conjecture* DC_n , posed for $n = 1$ as the first of six problems (Problème 11.1) of [1], asserts that every $\mathbb{C}$ -algebra endomorphism of A_n is an automorphism. Since A_n is simple, every endomorphism is injective; the content is surjectivity.

It is classical that DC_n implies the Jacobian conjecture JC_n [2, 8], and a celebrated converse of Tsuchimoto [3] and Belov-Kanel–Kontsevich [5] (see also Adjamağbo–van den Essen [7]) shows $JC_{2n} \Rightarrow DC_n$: the two conjectures are stably equivalent.

On July 19, 2026 (US time), Alpöge announced [9] an explicit polynomial map $F: \mathbb{C}^3 \rightarrow \mathbb{C}^3$ with constant Jacobian determinant -2 which is not injective, disproving JC_3 (hence JC_n for all $n \geq 3$). The announcement exhibits the explicit collision

$$(1) \quad F(0, 0, -\tfrac{1}{4}) = F(1, -\tfrac{3}{2}, \tfrac{13}{2}) = F(-1, \tfrac{3}{2}, \tfrac{13}{2}) = (-\tfrac{1}{4}, 0, 0),$$

which is machine-checked in our ancillary development (`alpöge_not_injective`; Section 4). By the contrapositive of $DC_3 \Rightarrow JC_3$, the *existence* of a non-surjective endomorphism of A_3 is an immediate consequence, and it is immediate to experts from the announcement. Existence, however, is where the immediate part stops. Read backwards, the implication $DC_3 \Rightarrow JC_3$ is *fully constructive* (unlike the char- p /ultraproduct direction of the stable equivalence), and the purpose of this note is to write the witness down, with machine-checked certification: an explicit strict embedding $A_3 \hookrightarrow A_3$.

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Prepared with assistance from AI systems (Claude Fable 5, Anthropic; cross-checked with GPT-5.6-sol, OpenAI). All computational claims are certified by the ancillary `sympy` script and by Lean 4/mathlib proofs; see Section 4.

Note added July 21. After a first version of this note was posted we learned of an independent note of Pickhardt, prepared with an AI research agent and dated July 19, 2026 [12], which carries out the same transfer for Alpöge's map, including an elegant proof that the pullback preserves differential order exactly. The construction is canonical and the overlap on the main theorem is complete; the Lean certification (Section 4) and the corollaries of Section 2 appear only in the present note.

Main result. Let $u = 1 + x_1x_2$ and let $F = (F_1, F_2, F_3)$ be Alpöge's map,

$$\begin{aligned} F_1 &= u^3x_3 + x_2^2u(4 + 3x_1x_2), \\ F_2 &= x_2 + 3x_1u^2x_3 + 3x_1x_2^2(4 + 3x_1x_2), \\ F_3 &= 2x_1 - 3x_1^2x_2 - x_1^3x_3, \end{aligned}$$

with Jacobian matrix $J = (\partial F_i / \partial x_j)$, $\det J = -2$ identically, and adjugate $M = \text{adj } J$, so that $JM = MJ = -2I_3$ with M a matrix of integer polynomials. Set $N = -\frac{1}{2}M = J^{-1}$.

Theorem 1.1. *The assignment*

$$\varphi(\hat{x}_i) = F_i(\hat{x}_1, \hat{x}_2, \hat{x}_3), \quad \varphi(\partial_i) = D_i := \sum_{k=1}^3 N_{ki}(\hat{x}_1, \hat{x}_2, \hat{x}_3) \partial_k$$

extends to a $\mathbb{C}$ -algebra endomorphism $\varphi: A_3 \rightarrow A_3$ which is injective and not surjective. In particular the Dixmier conjecture fails for A_3 , and it fails for A_n for every $n \geq 3$: under $A_n \cong A_3 \otimes A_{n-3}$ the endomorphism $\varphi \otimes \text{id}$ is injective with image $\varphi(A_3) \otimes A_{n-3}$, which is proper, since over a field $W \subsetneq V$ implies $W \otimes U \subsetneq V \otimes U$.

We emphasize that the lift $F \mapsto \varphi$ itself is *classical*: it is precisely the construction underlying the implication $\text{DC}_n \Rightarrow \text{JC}_n$, appearing already on p. 297 of [2] (where the Weyl-algebra question is attributed to communications of L. Vaserstein and V. Kac) and in [8, Ch. 10]; it is written out for arbitrary constant Jacobian determinant, with a classification of all such extensions, in [11] (see also [10]). What is new is the input, since until this month the recipe had no non-invertible Keller map to apply it to, together with the explicit inverse matrix N , the certified identities, and the resulting concrete witness. The reason everything is checkable is that all lifted operators are of order ≤ 1 in the ∂ 's, so the Weyl-algebra commutators reduce to commutators of vector fields and multiplication operators, with no ordering ambiguities or quantum corrections. Well-definedness is then *equivalent* to two families of polynomial identities:

- **(Inverse)** $JN = I_3$, giving $[D_i, F_j] = \sum_k N_{ki}(\partial F_j / \partial x_k) = (JN)_{ji} = \delta_{ij}$;
- **(Flatness)** for all i, j, l : $\sum_k (N_{ki} \partial_k N_{lj} - N_{kj} \partial_k N_{li}) = 0$, giving, after the symmetric second-order terms cancel, $[D_i, D_j] = \sum_l (\sum_k N_{ki} \partial_k N_{lj} - N_{kj} \partial_k N_{li}) \partial_l = 0$.

(The relations $[\varphi(\hat{x}_i), \varphi(\hat{x}_j)] = 0$ hold trivially.) Both families are certified by computer algebra and by Lean 4 proofs (Section 4). Abstractly, flatness holds because étale-locally $D_i = F^*(\partial / \partial y_i)$ is the pushforward of the commuting frame of coordinate vector fields under the local inverse of F ; the identities above are the global polynomial incarnation of this statement.

Proof of non-surjectivity. Injectivity is automatic: the kernel is a two-sided ideal and A_3 is simple. The subalgebra $\mathbb{C}[\hat{x}_1, \hat{x}_2, \hat{x}_3] \subset A_3$ is maximal commutative (if $T = \sum c_{\alpha\beta} \hat{x}^\alpha \partial^\beta$ commutes with every $\hat{x}_i$, then $[T, \hat{x}_i] = \sum \beta_i c_{\alpha\beta} \hat{x}^\alpha \partial^{\beta - e_i} = 0$ forces $c_{\alpha\beta} = 0$ whenever $\beta \neq 0$). Suppose φ were surjective, hence an automorphism. Then $\varphi(\mathbb{C}[\hat{x}]) = \mathbb{C}[F_1, F_2, F_3]$ would again be maximal commutative. But $\mathbb{C}[F] \subseteq \mathbb{C}[\hat{x}]$, and $\mathbb{C}[F] \neq \mathbb{C}[\hat{x}]$: an equality would give polynomials P_i with $x_i = P_i(F_1, F_2, F_3)$, so $P \circ F = \text{id}$ and F would be injective, contradicting the collision (1). Since $\mathbb{C}[\hat{x}]$ is commutative and contains $\mathbb{C}[F]$ strictly, $\mathbb{C}[F]$ is not maximal commutative, a contradiction. $\square$

2. THE POISSON COUNTEREXAMPLE AND A SYMPLECTIC KELLER MAP

Recall the Poisson conjecture PC_n [7]: every endomorphism of the polynomial algebra $\mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n]$ preserving the canonical Poisson bracket $\{x_i, y_j\} = \delta_{ij}$, $\{x_i, x_j\} = \{y_i, y_j\} = 0$, is an automorphism. The classical shadow of φ is the cotangent lift

$$\Phi: \mathbb{C}^6 \rightarrow \mathbb{C}^6, \quad \Phi(x, y) = (F(x), N(x)^\top y),$$

on $\mathbb{C}^6 = T^*\mathbb{C}^3$. (With the convention $[\partial_i, \hat{x}_j] = \delta_{ij}$ the symbol of ∂_i is $-y_i$; the two resulting sign changes cancel, so Φ is precisely the shadow of φ .)

Theorem 2.1. *Φ preserves the canonical Poisson bracket, satisfies $\det J\Phi = 1$ identically, and is not injective. In particular the pullback Φ^* is a non-invertible Poisson endomorphism, so the Poisson conjecture PC_3 fails, constructively, and Φ is a Keller counterexample on $\mathbb{C}^6$ with Jacobian determinant identically 1.*

Proof. Bracket preservation is the same pair of identity families as in Theorem 1.1 ($\{\Phi_i, \Phi_j\}$ in the three block cases reduces to (Inverse) and (Flatness)); $\det J\Phi = \det J \cdot \det N^\top = \det J / \det J = 1$ by the block-triangular structure (the y -block is linear in y with matrix $N^\top(x)$). For non-injectivity, note $N(x) \in \text{GL}_3(\mathbb{C})$ for every x , since $J(x)N(x) = I_3$ pointwise. Taking the collision (1), say $a \neq b$ with $F(a) = F(b)$, any y_a yields $y_b := (N(b)^\top)^{-1}N(a)^\top y_a$ with $\Phi(a, y_a) = \Phi(b, y_b)$ and $(a, y_a) \neq (b, y_b)$. $\square$

Remark 2.2. A unimodular Keller counterexample on $\mathbb{C}^3$ itself is immediate from Alpöge's map by a linear volume correction (compose with $\text{diag}(-\frac{1}{2}, 1, 1)$). The point of Φ is that it is *symplectic*, which no map on odd-dimensional $\mathbb{C}^3$ can be; to our knowledge it is the first symplectic Keller counterexample. A resultant computation, not needed for any result here, shows F (hence Φ) has geometric degree exactly 3. Note PC_1 is essentially verbatim JC_2 , which remains open. Finally, PC_n fails for every $n \geq 3$: extending Φ by the identity on the remaining $n - 3$ conjugate pairs preserves the canonical bracket, the block determinant, and the collision of Theorem 2.1.

3. THE CONTACT COUNTEREXAMPLE

The lift Φ preserves more than the symplectic form: it preserves the Liouville form itself. Writing $\lambda = \sum_i y_i dx_i$ on $\mathbb{C}^6$,

$$\Phi^*\lambda = \sum_i (N^\top y)_i dF_i = \sum_k ((NJ)^\top y)_k dx_k = \lambda,$$

since $NJ = I_3$ (certified; Section 4). Consider $\mathbb{C}^7$ with coordinates (x, y, z) and the standard contact form $\alpha = dz - \lambda$. Since no contact analogue of the Jacobian conjecture appears in the literature, we first state the natural one.

Contact Jacobian question. Is every polynomial endomorphism of $\mathbb{C}^{2n+1}$ that preserves the standard contact form α exactly (a polynomial strict contactomorphism) an automorphism?

Theorem 3.1. *The map $\Psi: \mathbb{C}^7 \rightarrow \mathbb{C}^7$, $\Psi(x, y, z) = (F(x), N(x)^\top y, z)$, satisfies $\Psi^*\alpha = \alpha$, $\det J\Psi = 1$ identically, and is not injective. Hence the contact Jacobian question has a negative answer on $\mathbb{C}^{2n+1}$ for every $n \geq 3$.*

Proof. $\Psi^*\alpha = dz - \Phi^*\lambda = dz - \lambda = \alpha$ by the displayed identity, and $\det J\Psi = \det J\Phi \cdot 1 = 1$ by Theorem 2.1. Non-injectivity is inherited from Φ with any fixed z . Stabilization adds conjugate pairs to the (x, y) -block as in Theorem 2.1; the contact form on $\mathbb{C}^{2n+1}$ restricts compatibly. $\square$

Remark 3.2. The frontier cases are open: a strict-contact counterexample on $\mathbb{C}^5$ would require a symplectic Keller counterexample on $\mathbb{C}^4$, i.e. the PC_2 territory, and $\mathbb{C}^3$ likewise reduces to the plane. Preservation of α exactly is the strongest form; we do not know whether the weaker conformal-contact question (preserving α up to a polynomial multiplier) admits smaller counterexamples.

4. CERTIFICATION

All computational claims underlying Theorems 1.1 and 2.1 reduce to polynomial identities over $\mathbb{Q}$ in three (resp. six) variables, plus three rational point evaluations (the collision (1)), and are certified twice, independently:

- (1) **Computer algebra.** The ancillary file `ancillary_check.py` (sympy, exact rational arithmetic) verifies $\det J = -2$, $JN = I_3$, the nine flatness identities, bracket preservation for Φ , and $\det J\Phi = 1$.
- (2) **Machine-checked proofs.** A Lean 4/mathlib development (ancillary) proves, with no `sorry` and no axioms beyond the standard three: the Keller property $\det J = -2$ with the Jacobian entries certified as formal partial derivatives (`MvPolynomial.pderiv`); the collision (1) (`alpoge_not_injective`); the identities $J \cdot \text{adj } J = \text{adj } J \cdot J = -2 I_3$ (`dixmier_inverse`, `dixmier_inverse'`); the nine adjugate entries as explicit polynomials; the full quantified flatness statement $\sum_k ((\text{adj } J)_{ki} \partial_k (\text{adj } J)_{lj} - (\text{adj } J)_{kj} \partial_k (\text{adj } J)_{li}) = 0$ for all i, j, l (`dixmier_flatness`, all 27 cases); and the scalar-cleared determinant of the block map, $\det \begin{pmatrix} J & 0 \\ C & (\text{adj } J)^\top \end{pmatrix} = -8 = (-2)^3 \cdot 1$ for *every* lower-left block C (`poisson_det_scaled`), certifying $\det J\Phi = 1$; the identity $NJ = I_3$ used in Section 3 is `dixmier_inverse'` after the same scalar clearing. The Lean statements are the scalar-cleared integral forms; dividing by the unit -2 (resp. $(-2)^2$, $(-2)^3$) recovers the identities as stated in the text, since $N = -\frac{1}{2} \text{adj } J$.

5. REMARKS ON $n \leq 2$

DC_1 and DC_2 remain open. Since $\text{DC}_2 \Rightarrow \text{JC}_2$, the two-variable Dixmier conjecture is at least as strong as the plane Jacobian conjecture. In the opposite direction, $\text{JC}_2 \Rightarrow \text{DC}_1$ [5]; a recent preprint of Zheglov [13] proposes a proof of DC_1 unconditionally. Our construction sharpens the $n = 1$ landscape as follows. Suppose DC_1 holds. Then no endomorphism of A_1 can have a plane Keller counterexample as its classical shadow, in any formalization of “shadow” under which automorphisms of A_1 induce polynomial symplectomorphisms of $\mathbb{C}^2$ [6, 4]. By contrast the cotangent lift, being linear in the momenta, quantizes with no corrections in every dimension (Theorem 1.1). Making the one- and two-variable obstruction precise is, we believe, a fruitful new question about the plane Jacobian problem.

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